Topic 5: Alkenes

Related topics in Units 2, 4 and 5 will assume knowledge of this material.

Students will be assessed on their ability to:

5.1	know the general formula of alkenes and understand that alkenes and cycloalkenes are hydrocarbons which are unsaturated (have a carbon-carbon double bond which consists of a σ bond and a π bond)
5.2	be able to explain geometric isomerism in terms of restricted rotation around a C=C double bond and the nature of the substituents on the carbon atoms
5.3	understand the *E–Z* naming system for geometric isomers and why it is necessary to use this when the *cis*- and *trans*- naming system breaks down
5.4	be able to describe the reactions of alkenes, limited to: i the addition of hydrogen, using a nickel catalyst, to form an alkane ii the addition of halogens to produce a di-substituted halogenoalkane iii the addition of hydrogen halides to produce mono-substituted halogenoalkanes iv the addition of steam, in the presence of an acid catalyst, to produce alcohols v oxidation of the double bond by acidified potassium manganate(VII) to produce a diol
5.5	know the qualitative test for a C=C double bond using bromine or bromine water
5.6	be able to describe the mechanism (including diagrams), giving evidence where possible, of: i the electrophilic addition of bromine and hydrogen bromide to ethene ii the electrophilic addition of hydrogen bromide to propene *Use of the curly arrow notation is expected – the curly arrows should start from either a bond or from a lone pair of electrons.* *Knowledge of the relative stability of primary, secondary and tertiary carbocation intermediates is expected.*
5.7	be able to describe the addition polymerisation of alkenes and draw the repeat unit given the monomer, and vice versa
5.8	understand how chemists limit the problems caused by polymer disposal by: i developing biodegradable polymers ii removing toxic waste gases produced by the incineration of polymers
	Further suggested practicals: i investigating the difference in reactivity of alkanes and alkenes, including combustion, reaction with bromine water, reaction with acidified potassium manganate(VII) ii preparation of cyclohexene from cyclohexanol iii preparation of limonene from orange peel by steam distillation iv preparation of Perspex® from methyl 2-methylpropenoate

Unit 2: Energetics, Group Chemistry, Halogenoalkanes and Alcohols

IAS compulsory unit

Externally assessed

Unit description

Introduction	This unit develops the treatment of chemical bonding by introducing intermediate types of bonding and by exploring the nature and effects of intermolecular forces. Study of the Periodic Table is extended to cover the chemistry of Groups 1, 2 and 7, where ideas about redox reactions are applied to the reactions of halogens and their compounds. The study of energetics in chemistry is of theoretical and practical importance. In this unit, students learn how to define, measure and calculate enthalpy changes. They will see how a study of enthalpy changes helps chemists to understand chemical bonding. The unit also develops an understanding – mostly at a qualitative level – of the ways in which chemists can control the rate, direction and extent of chemical change in reactions. The organic chemistry in this unit covers halogenoalkanes and alcohols, and explores the mechanisms of selected reactions. The study of spectroscopy gives further examples of the importance of accurate and sensitive methods of analysis, which can be applied to study chemical changes but also to detect drugs such as ethanol.
Chemistry in action	The use of models in chemistry is illustrated by the way in which the Maxwell-Boltzmann distribution and collision theory can account for the effects of temperature on the rates of chemical reactions. The unit shows how chemists can study chemical changes at an atomic level and propose mechanisms to account for their observations. The study of rates and equilibria shows the contribution that chemistry can make to a more sustainable economy by redeveloping manufacturing processes to make them more efficient, less hazardous and less polluting.

Practical skills	Again, students can begin their practical work for this unit with simple reactions, in polystyrene cups, to investigate energy changes in chemical reactions. This leads to the first core practical in this unit. Inorganic chemistry and basic redox reactions can also be explored with simple test-tube reactions. The techniques of volumetric analysis are introduced in this unit, with two core practicals to develop competence in this key skill. The effect of temperature, concentration and surface area on the rate of a reaction can be explored through a variety of reactions, and also forms part of the first core practical in organic chemistry. Further core practicals in organic chemistry are used to develop students' skills in using glassware and techniques such as reflux, use of a separating funnel and distillation. The final core practical considers qualitative analysis for ions and organic functional groups.
Mathematical skills	There are opportunities for the development of mathematical skills in this unit. This includes plotting and extrapolating graphs of temperature rise against time for displacement reactions, calculating enthalpy changes in J and kJ mol^{-1}, using algebra to solve Hess's Law problems, calculating enthalpy changes using bond enthalpies and evaluating experimental results in terms of measurement uncertainties and systematic errors in the context of measuring energy changes and in titrations. Also, calculating oxidation numbers within a complex system, balancing equations for redox reactions by combining ionic half-equations, calculating rates from reaction time, plotting graphs and having an appreciation of the graph for a Maxwell-Boltzmann distribution, and deriving an algebraic expression for the equilibrium constant. Further mathematical skills can be developed in analysing mass and infrared spectra. (Please see *Appendix 6: Mathematical skills and exemplifications* for more details.)

Assessment information

- First assessment: June 2019.
- The assessment is 1 hour and 30 minutes.
- The assessment is out of 80 marks.
- Students must answer all questions.
- This paper has three sections:
    - Section A: multiple choice questions
    - Section B: mixture of short-open, open-response, calculations and extended-writing questions
    - Section C: contemporary context question.
- This paper will contain questions that require information from the Data Booklet (see *Appendix 9*).
- This paper will include a minimum of 18 marks that target mathematics at Level 2 or above.
- Students will be expected to apply their knowledge and understanding of experimental methods in familiar and unfamiliar contexts.
- This paper may contain some synoptic questions which require knowledge and understanding from Unit 1.
- Calculators may be used in the examination (see *Appendix 8: Use of calculators*).